CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications

Course: MCA412-2 - Research Methodology

LAB Exercise II –

Research Gap Identification.

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Name : Aditya Kumar

Reg. No : 2647203
Class : 2 MCA – B

Domain :

Marine Ecological Informatics & Bio-Optical Remote Sensing

Lab 2

Lab Exercise II:
Research Gap Identification.

1. Literature Summary and Findings

The five research papers reviewed in Lab 1 establish the current technical capabilities and physical limitations of monitoring marine ecosystems, harmful algal blooms, and ocean optics using satellite platforms and environmental datasets. Pandi et al. (2020) demonstrated through in-situ hyperspectral radiometry that the ratio Rrs(531)/Rrs(510) provides a direct optical proxy for Noctiluca scintillans cell density, although their empirical model relies heavily on localized sampling. Shi and Wang (2022) identified the different physical drivers of seasonal Arabian Sea phytoplankton blooms: summer blooms are associated with wind-driven coastal upwelling and vertical velocities exceeding 1 m/day, while winter blooms are linked to convective mixing and mixed-layer deepening to about 80 m. Kuttippurath et al. (2023) connected atmospheric dynamics with marine biology, reporting a long-term decline in primary productivity associated with surface warming and showing that changes in dust transport can affect nutrient inputs into the northern Indian Ocean. Li et al. (2024) examined satellite data quality and found that standard atmospheric correction methods lose substantial accuracy under high solar zenith angles (>70°), while a neural-network approach improved retrieval accuracy. Finally, Cael et al. (2023) showed that treating multi-band remote-sensing reflectance as a multivariate signal, while accounting for autocorrelation, can reveal climate-related ecological changes that may be obscured when using chlorophyll alone.

Overall, the literature indicates that satellite ocean-colour monitoring has advanced considerably, but several challenges remain. These include spectral confusion when identifying mixotrophic blooms, the limited connection between surface observations and subsurface physical and biogeochemical processes, the influence of atmospheric dust on nutrient availability, reduced data quality under difficult viewing conditions, and the difficulty of separating genuine ecological trends from natural variability and statistical noise.

2. Five Issues Identified in the Existing Literature

Issue 1: Spectral ambiguity and misclassification in empirical chlorophyll algorithms during mixotrophic blooms

What is missing / could be improved: Standard satellite ocean-colour models, including common blue-green band-ratio algorithms, are mainly designed for typical phytoplankton conditions and may not accurately identify Noctiluca scintillans. Its photosynthetic endosymbionts contribute distinctive optical features, including absorption near 510 nm and a reflectance peak near 531 nm, which can affect biomass estimates.

New questions that arise: Can a multi-band ratio approach using the 510 nm and 531 nm regions reliably distinguish Noctiluca blooms from other phytoplankton populations across broad satellite coverage?

How my research could contribute: I can develop a feature-engineering workflow using diagnostic reflectance ratios, such as Rrs(531)/Rrs(510), from MODIS and VIIRS Level-2 data to explore an automated bloom-classification index.

Future direction suggested by the authors: Pandi et al. (2020) point to the value of using the diagnostic spectral bands and region-specific bio-optical tuning to improve the identification of dinoflagellate blooms.

Issue 2: Decoupling of satellite surface observations from vertical hydrodynamic and mixing processes

What is missing / could be improved: Remote-sensing studies often treat surface optical images as two-dimensional snapshots. This can leave out vertical processes such as mixed-layer deepening, upwelling, and subsurface circulation that influence bloom development and persistence.

New questions that arise: How strongly are local changes in mixed-layer depth (MLD) and vertical velocity (w) associated with the onset and expansion of winter bloom events in the northern Arabian Sea?

How my research could contribute: I can compare satellite reflectance data with ocean reanalysis variables from sources such as Copernicus Marine (CMEMS) and GODAS, including sea-surface temperature, mixed-layer depth, and vertical velocity, to investigate possible environmental triggers.

Future direction suggested by the authors: Shi and Wang (2022) show the importance of considering hydrographic conditions and vertical movement alongside satellite observations when explaining Arabian Sea phytoplankton dynamics.

Issue 3: Limited consideration of atmospheric dust deposition and short-term meteorological events

What is missing / could be improved: Marine ecosystem studies may focus mainly on ocean circulation when explaining nutrient supply. However, dust transported from nearby arid regions can add minerals, including iron and phosphorus, and may influence blooms even when coastal upwelling is weak.

New questions that arise: Can aerosol optical depth (AOD) and changes in dust-transport pathways help indicate or explain occasional algal blooms in stratified open-ocean waters?

How my research could contribute: I can include atmospheric reanalysis variables, such as MERRA-2 aerosol and dust measures, in the exploratory data analysis pipeline and examine their relationship with sudden changes in chlorophyll.

Future direction suggested by the authors: Kuttippurath et al. (2023) highlight the importance of studying atmospheric and oceanic conditions together to distinguish possible dust-related fertilisation from the effects of normal coastal upwelling.

Issue 4: Reduced atmospheric-correction accuracy under extreme solar viewing geometry

What is missing / could be improved: Satellite reflectance, particularly in blue wavelengths such as 412 nm and 443 nm, can be affected by substantial errors when solar zenith angles are high (>70°) or viewing angles are large. This can lead to missing observations, especially during winter.

New questions that arise: Can machine-learning or neural-network atmospheric-correction methods improve reflectance retrieval under high solar zenith angles without introducing persistent spectral bias?

How my research could contribute: I can apply quality and geometry checks and investigate neural-network correction approaches, such as NN-V, to understand whether the amount of usable winter satellite data can be improved.

Future direction suggested by the authors: Li et al. (2024) demonstrate the potential of advanced neural-network atmospheric correction for improving ocean-colour retrievals under challenging illumination conditions.

Issue 5: Long-term ecological trends obscured by univariate noise and autocorrelation

What is missing / could be improved: Long-term monitoring often relies on chlorophyll-a as a single indicator. Natural year-to-year variation can make trends difficult to identify, while failing to account for temporal autocorrelation can make statistical results appear more significant than they are.

New questions that arise: Can multivariate analysis of visible-band reflectance, combined with autoregressive correction, reveal regional ecological changes earlier or more reliably than chlorophyll-only trend analysis?

How my research could contribute: I can test a multivariate regression workflow with Cochrane-Orcutt autocorrelation correction on multi-band satellite data to investigate ecological trends over an approximately 20-year observation period.

Future direction suggested by the authors: Cael et al. (2023) support the use of multi-band optical information and careful statistical treatment to identify subtle ecological changes in the surface ocean.